

Lab 1 Z-transfer Functions, Difference Equations, and Filter Implementation Using MATLAB

OBJECTIVES

1. To learn how to determine the difference equation given FIR (finite-impulse response) system coefficients.
2. To learn how to determine the FIR transfer function based on the given difference equation, and learn how to calculate and display frequency responses of the FIR system and perform digital filtering.

PROCEDURE:

Part A: FIR Filters

An FIR system is given below:

Numerator coefficients: $b=[0.2 \ 0.3 \ 0.2]$

- a. Determine its transfer function $H(z)=$
- b. Determine its difference equation $y(n)=$
- c. Given sampled values: $\text{sample}=[1:1:10]$, write a MATLAB program to perform the digital filtering:

```
-----  
    sample =[1:1:10]  
    x=[0 0 sample]  
    for n=3:1:length(x)  
        y(n)= completed on your own  
    end  
    y(3:1:length(x))  
-----
```

Using the standard MATLAB routine “filter()” for verification:
 $\text{yy} = \text{filter}([0.2 \ 0.3 \ 0.2], 1, \text{sample})$

Is the vector containing $y(3:1:\text{length}(\text{sample}))$ same as the vector yy ? _____

Two FIR system coefficients are given below, respectively,

FIR system 1:

$\text{bLP} = [$
-0.0012 -0.0025 -0.0045 -0.0068 -0.0073 -0.0030 0.0089 ...
0.0297 0.0583 0.0907 0.1208 0.1422 0.1500 0.1422 ...
0.1208 0.0907 0.0583 0.0297 0.0089 -0.0030 -0.0073 ...
-0.0068 -0.0045 -0.0025 -0.0012]

FIR system 2:

$\text{bBP} = [$
0.0004 -0.0017 -0.0064 -0.0076 0.0073 0.0363 0.0458 ...
0.0000 -0.0802 -0.1134 -0.0419 0.0860 0.1500 0.0860 ...
-0.0419 -0.1134 -0.0802 0.0000 0.0458 0.0363 0.0073 ...
-0.0076 -0.0064 -0.0017 0.0004]

1. Determine the transfer function using z-transform for the first FIR system.
2. Determine the difference equation for the first FIR system.

Plot the frequency responses using the MATLAB function `freqz()` for each of z-transfer functions.

MATLAB>>freqz(bLP,1,512,8000); % sampling rate=8000 Hz

Label and print your graph.

Perform similar operations for FIR systems with coefficients: bBP and label and print your plots.

3. Generate the sum of sinusoids for duration of 1 second:

$x(t) = 5 \cos(2\pi(500)t) + 5 \cos(2\pi(1200)t + 0.25\pi) + 5 \cos(2\pi(1800)t + 0.5\pi)$ using a sampling rate of 8 kHz.

4. Write MATLAB programs using the MATLAB function **filter()** according to the two given FIR systems.

- a. Perform digital filtering using each of the developed programs, plot the input signal spectrum and the output signal spectrum using the frequency resolution of 1 Hz, respectively.

Print and label the spectrum.

What frequency components/component (frequency locations of the spectral peak(s)) are/is shown in the output signal spectrum?

- b. Listen to and compare the input signal and output signal from each filter, respectively. Discuss the obtained results.